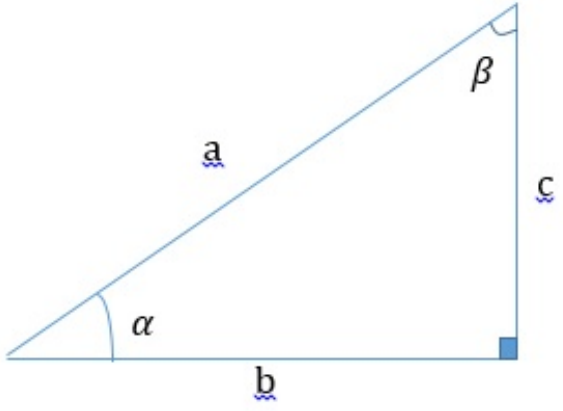
Trigonometria

Docente: Andreia Monteiro
Matemática Computacional

Trigonometria no triângulo retângulo

Seja $[ABC]$ um triângulo retângulo. Então, as razões trigonométricas são dadas por:

$$seno = \frac{cateto\ oposto}{hipotenusa} \qquad coseno = \frac{cateto\ adjacente}{hipotenusa} \qquad tangente = \frac{cateto\ oposto}{cateto\ adjacente} \qquad cotangente = \frac{cateto\ adjacente}{cateto\ oposto}$$



$$sen(\alpha) = \frac{c}{a} \qquad cos(\alpha) = \frac{b}{a}$$

$$tg(\alpha) = \frac{c}{b} \qquad cotg(\alpha) = \frac{b}{c}$$

Fórmulas Trigonométicas do mesmo ângulo

- $\sin^2\alpha + \cos^2\alpha = 1$ Fórmula Fundamental da Trigonometria
- $\tan\alpha = \frac{\sin\alpha}{\cos\alpha}$
- $\cot\alpha = \frac{\cos\alpha}{\sin\alpha}$
- $\tan^2\alpha + 1 = \frac{1}{\cos^2\alpha}$
- $\cot^2\alpha + 1 = \frac{1}{\sin^2\alpha}$

Soma e Diferença

Sendo a e b dois ângulos dados, então:

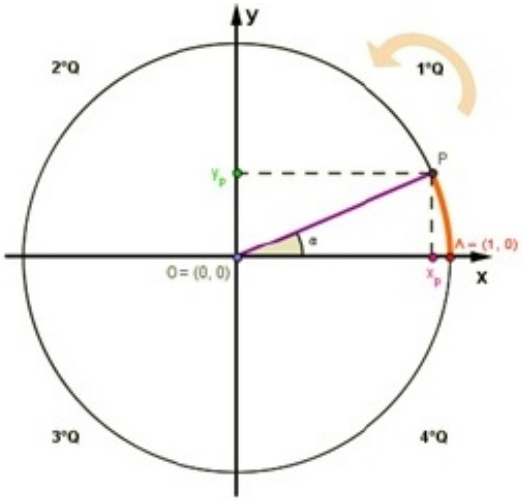
$$\begin{aligned}
\operatorname{sen}(A \pm B) &= \operatorname{sen}A \cos B \pm \operatorname{sen}B \cos A \\
\cos(A \pm B) &= \cos A \cos B \mp \operatorname{sen}A \operatorname{sen}B \\
\tan(A \pm B) &= \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B} \\
\cot(A \pm B) &= \frac{\cot A \cot B \mp 1}{\cot B \pm \cot A}
\end{aligned}$$

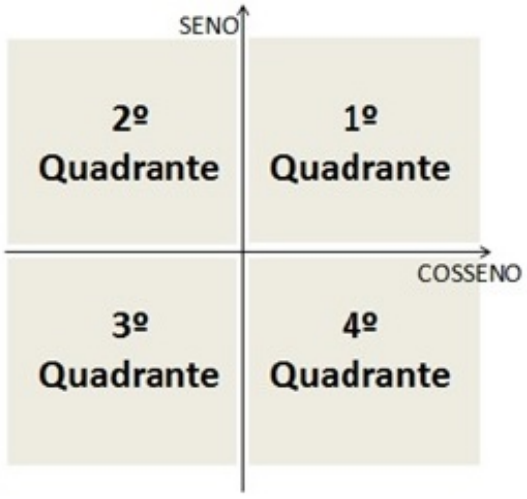
Fórmulas de Duplicação

Sendo *a* um ângulo, então:

$$\sin(2a) = 2\sin a\cos a$$
$$\cos(2a) = \cos^2 a - \sin^2 a$$
$$\tan(2a) = \frac{2\tan a}{1-\tan^2 a}(a \neq \frac{\pi}{2} + k\pi, a \neq \frac{\pi}{4} + \frac{k\pi}{2}), k \in \mathbb{Z}$$
$$\cot(2a) = \frac{\cot^2 a - 1}{2\cot a}(a \neq \frac{k\pi}{2}, k \in \mathbb{Z})$$

Razões Trigonométricas





	sin	cos	tan	cot
1.ºQ	+	+	+	+
2.ºQ	+	-	-	-
3.ºQ	-	-	+	+
4.ºQ	-	+	-	-

	sin	cos	tan	cot
0	0	1	0	ND
$\frac{\pi}{2}$	1	0	ND	0
π	0	-1	0	ND
$\frac{3\pi}{2}$	-1	0	ND	0

	sin	cos	tan	cot
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	$\sqrt{3}$
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{\sqrt{3}}{3}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	1

Redução ao Primeiro Quadrante

Sendo $\alpha \in 1Q$, então:

$\sin(\pi - \alpha) = \sin\alpha$ $\cos(\pi - \alpha) = -\cos\alpha$ $\tan(\pi - \alpha) = -\tan\alpha$ $\cot(\pi - \alpha) = -\cot\alpha$	$\sin(\pi + \alpha) = -\sin\alpha$ $\cos(\pi + \alpha) = -\cos\alpha$ $\tan(\pi + \alpha) = \tan\alpha$ $\cot(\pi + \alpha) = \cot\alpha$
$\sin(\frac{\pi}{2} - \alpha) = \cos\alpha$ $\cos(\frac{\pi}{2} - \alpha) = \sin\alpha$ $\tan(\frac{\pi}{2} - \alpha) = \cot\alpha$ $\cot(\frac{\pi}{2} - \alpha) = \tan\alpha$	$\sin(\frac{\pi}{2} + \alpha) = \cos\alpha$ $\cos(\frac{\pi}{2} + \alpha) = -\sin\alpha$ $\tan(\frac{\pi}{2} + \alpha) = -\cot\alpha$ $\cot(\frac{\pi}{2} + \alpha) = -\tan\alpha$
$\sin(\frac{3\pi}{2} - \alpha) = -\cos\alpha$ $\cos(\frac{3\pi}{2} - \alpha) = -\sin\alpha$ $\tan(\frac{3\pi}{2} - \alpha) = \cot\alpha$ $\cot(\frac{3\pi}{2} - \alpha) = \tan\alpha$	$\sin(\frac{3\pi}{2} + \alpha) = -\cos\alpha$ $\cos(\frac{3\pi}{2} + \alpha) = \sin\alpha$ $\tan(\frac{3\pi}{2} + \alpha) = -\cot\alpha$ $\cot(\frac{3\pi}{2} + \alpha) = -\tan\alpha$

Equações Trigonométricas

Quando $p \in]-1;1[$, então a equação $\sin x = \sin p$ tem duas soluções em $[0;2\pi]$ e, sendo uma delas α, temos:

$$\sin x = \sin \alpha \Leftrightarrow x = \alpha + 2k\pi \ \vee \ x = \pi - \alpha + 2k\pi, \ k \in \mathbb{Z}$$

Casos particulares em que $p \in \{-1, 0, 1\}$

- $\sin x = -1 \Leftrightarrow x = -\frac{\pi}{2} + 2k\pi, \ k \in \mathbb{Z}$
- $\sin x = 0 \Leftrightarrow x = k\pi, \ k \in \mathbb{Z}$
- $\sin x = 1 \Leftrightarrow x = \frac{\pi}{2} + 2k\pi, \ k \in \mathbb{Z}$

Equações Trigonométricas

Quando $p \in]-1;1[$, então a equação $\cos x = \cos p$ também tem duas soluções em $[0;2\pi]$ e, sendo uma delas α, temos:

$$\cos x = \cos \alpha \Leftrightarrow x = \alpha + 2k\pi \ \vee \ x = -\alpha + 2k\pi, \ k \in \mathbb{Z}$$

Casos particulares em que $p \in \{-1, 0, 1\}$

- $\cos x = -1 \Leftrightarrow x = \pi + 2k\pi, \ k \in \mathbb{Z}$
- $\cos x = 0 \Leftrightarrow x = \frac{\pi}{2} + k\pi, \ k \in \mathbb{Z}$
- $\cos x = 1 \Leftrightarrow x = 2k\pi, \ k \in \mathbb{Z}$

Equações Trigonométricas

As equações do tipo $\tan x = p$ ou do tipo $\cot x = p$, têm sempre solução, qualquer que seja o valor de p. Recorde-se, além disso, que o período das funções tangente e cotangente é π, pelo que:

$$\tan x = \tan \alpha \Leftrightarrow x = \alpha + k\pi, \ \ k \in \mathbb{Z}$$

$$\cot x = \cot \alpha \Leftrightarrow x = \alpha + k\pi, \ \ k \in \mathbb{Z}$$

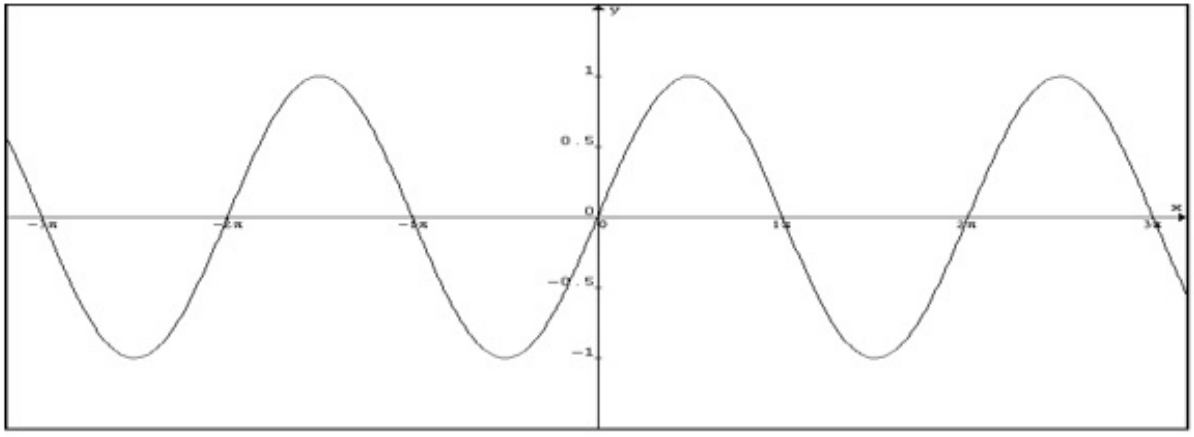
Observação

As equações do tipo $\tan x = p$ e $\cot x = p$ são equações fracionárias, de modo que, é necessário ter em atenção se as soluções encontradas pertencem ao domínio da respetiva equação.

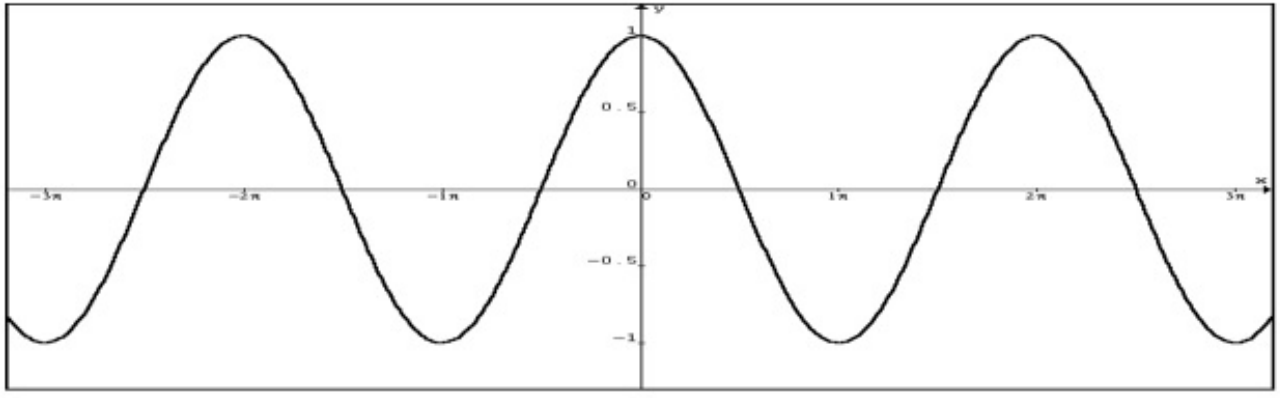
Funções Trigonométricas Inversas

As funções $\sin x$; $\cos x$; $\tan x$ e $\cot x$ não são injetivas e, como tal, não admitem inversa. Contudo, podemos considerar sempre uma restrição bijetiva, de cada uma das funções.

Sen x

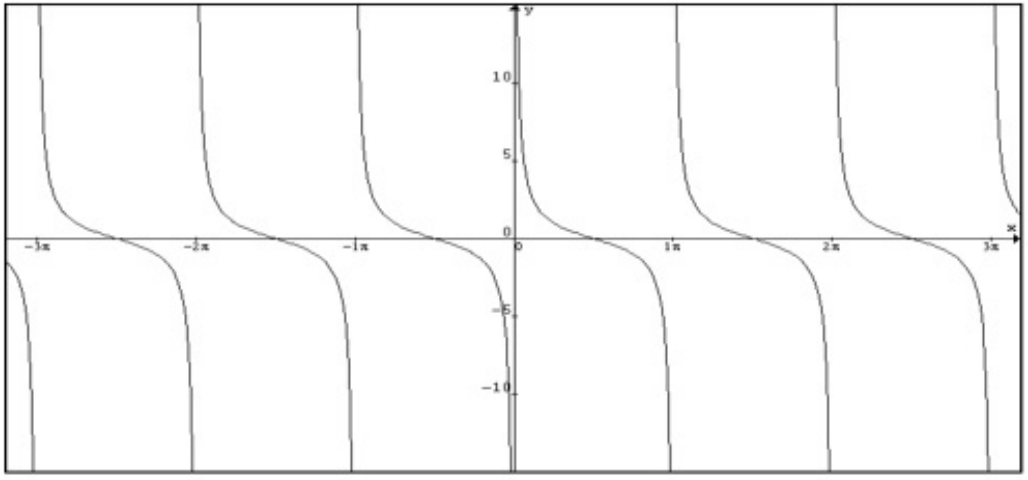


Cos x

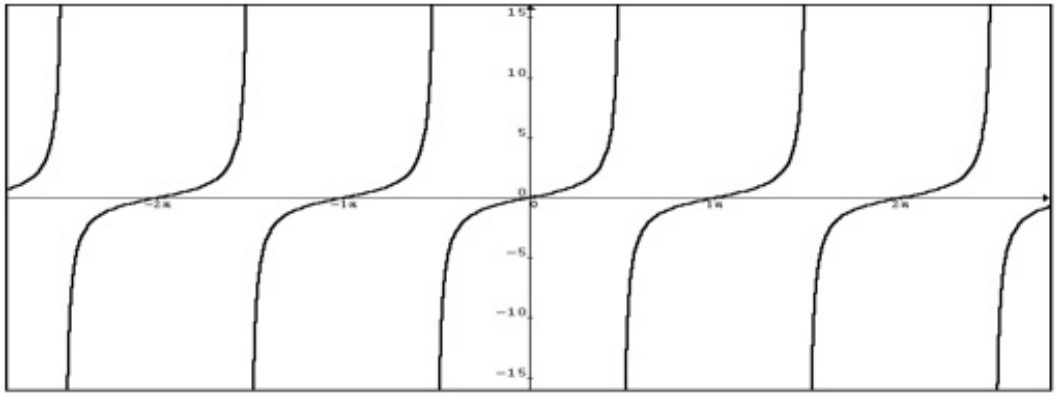


Funções Trigonométricas Inversas

Tan x



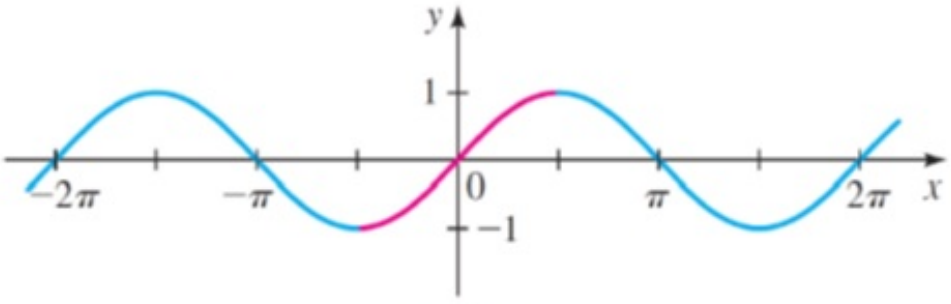
Cotg x



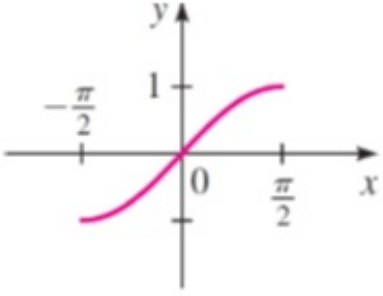
Funções Trigonométricas Inversas

A inversa da função seno designa-se por arco seno e define-se por:

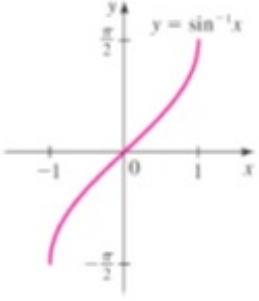
$$f^{-1} : [-1, 1] \longrightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$
$$x \longmapsto \arcsin x$$



$y = \sin x$



$y = \sin x, -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

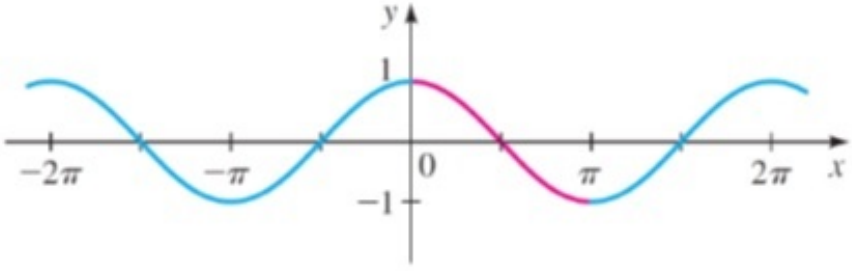


$\sin^{-1} x$

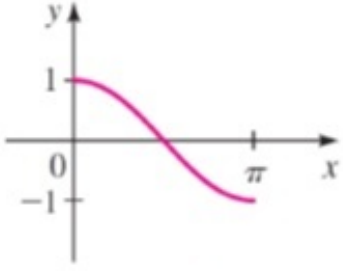
Funções Trigonométricas Inversas

A inversa da função cosseno designa-se por arco cosseno e define-se por:

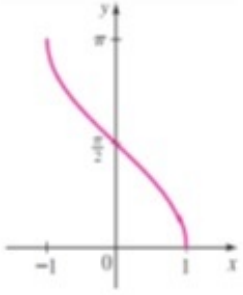
$$\begin{aligned} f^{-1} : [-1,1] &\longrightarrow [0,\pi] \\ x &\longmapsto \arccos x \end{aligned}$$



$y = \cos x$



$y = \cos x,\ 0 \leq x \leq \pi$

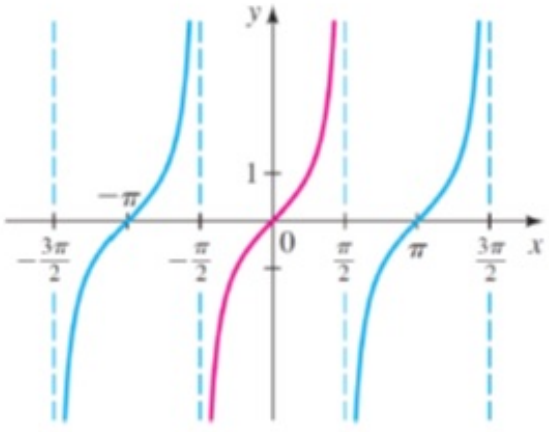


$\cos^{-1} x$

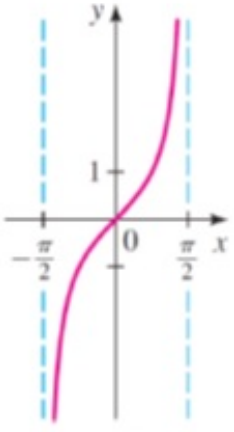
Funções Trigonométricas Inversas

A inversa da função tangente designa-se por arco tangente e define-se por:

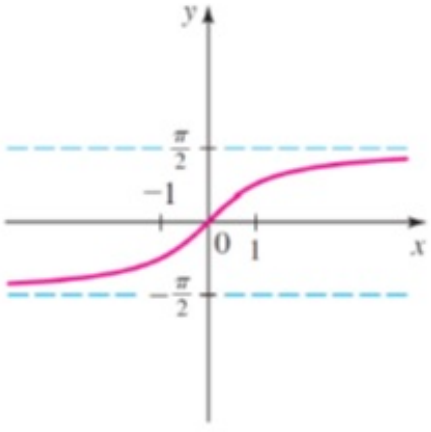
$$f^{-1} : \mathbb{R} \longrightarrow \left]-\frac{\pi}{2}, \frac{\pi}{2}\right[$$
$$x \longmapsto \arctan x$$



$y = \tan x$



$y = \tan x, \ -\frac{\pi}{2} < x < \frac{\pi}{2}$

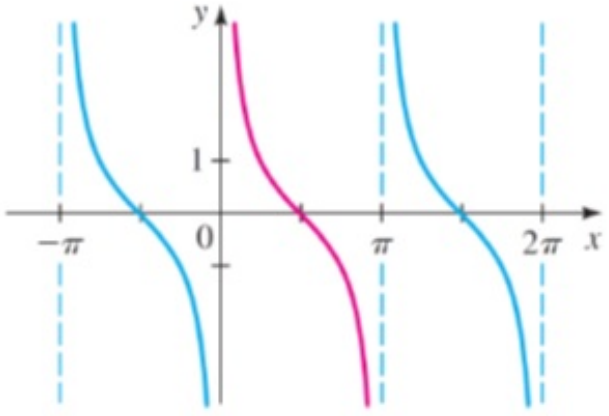


$y = \tan^{-1} x$

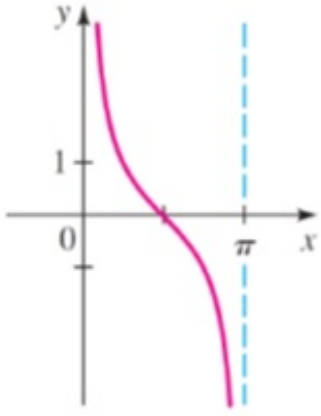
Funções Trigonométricas Inversas

A inversa da função cotangente designa-se por arco cotangente e define-se por:

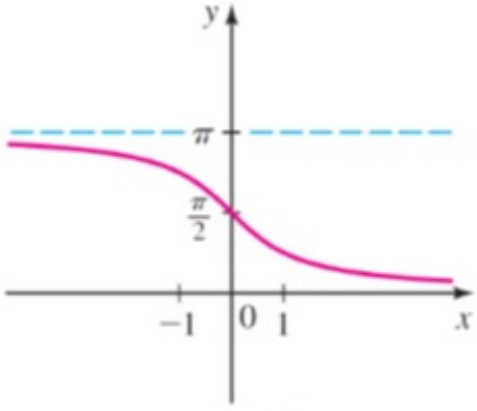
$$f : IR \to [0, \pi]$$
$$x \longmapsto arccot\, x$$



$y = \cot x$



$y = \cot x,\ 0 < x < \pi$



$y = \cot^{-1} x$

Alguns limites trigonométricos

- $\lim_{u \to 0} \frac{\sin u}{u} = 1$
- $\lim_{u \to 0} \frac{\tan u}{u} = 1$
- $\lim_{u \to 0} \frac{1 - \cos u}{u} = 0$